

End-to-End Fidelity Analysis of Quantum Circuit Optimization: From Gate-Level Transformations to Pulse-Level Control

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Abstract—We present a comprehensive analysis of quantum circuit fidelity across the full compilation stack, from high-level gate optimization through pulse-level control synthesis. Using a modular integration framework connecting a C++ circuit optimizer with Lindblad-based pulse simulation, we systematically evaluate the fidelity impact of four optimization passes—gate cancellation, commutation, rotation merging, and identity elimination—on IQM Garnet hardware parameters. Our simulation campaign spanning 371 circuit runs reveals that gate cancellation provides the most significant improvement (68% of circuits improved, 14,024 gates eliminated), while pulse duration exhibits the strongest negative correlation with process fidelity ($r = -0.74$, $R^2 = 0.55$). We validate these findings through hardware execution on the IQM Resonance Garnet 20-qubit processor, demonstrating 70% gate reduction on QFT circuits with 100% job success rate. Our open-source framework enables reproducible benchmarking of quantum compilation pipelines.

Index Terms—Quantum circuit optimization, quantum compilation, pulse-level control, Lindblad simulation, superconducting qubits, NISQ

I. INTRODUCTION

THE fidelity of quantum computations on near-term noisy intermediate-scale quantum (NISQ) devices [1] is fundamentally limited by gate errors, decoherence, and the overhead introduced during circuit compilation. While significant effort has been invested in developing gate-level optimization passes [2]–[4], the end-to-end impact of these transformations on physical pulse-level fidelity remains insufficiently characterized.

Modern quantum compilation involves multiple stages: (1) logical circuit optimization to reduce gate count and depth, (2) routing and mapping to hardware connectivity constraints, (3) native gate decomposition, and (4) pulse-level synthesis [5]. Each stage introduces potential fidelity degradation, yet optimization strategies are typically evaluated in isolation without considering downstream effects.

In this work, we present `qco-integration`, an end-to-end framework for analyzing quantum circuit fidelity across the full compilation stack. Our contributions include:

- A modular integration architecture connecting C++ circuit optimization with Python pulse synthesis
- Systematic evaluation of four optimization passes on 371 benchmark circuits

- Quantitative analysis of fidelity scaling with circuit parameters
- Hardware validation on IQM Resonance Garnet (20 qubits)
- Open-source tools for reproducible quantum compilation benchmarking

II. BACKGROUND

A. Circuit Optimization Passes

Gate-level optimization passes transform quantum circuits to reduce gate count and depth while preserving the unitary operation:

Gate Cancellation: Identifies and removes adjacent inverse gate pairs (e.g., $XX^\dagger = I$, $HH = I$).

Commutation Analysis: Propagates gates through each other when they commute ($[A, B] = 0$), enabling opportunities for subsequent cancellation.

Rotation Merging: Combines consecutive rotations about the same axis: $R_z(\theta_1)R_z(\theta_2) = R_z(\theta_1 + \theta_2)$.

Identity Elimination: Removes rotations that are multiples of 2π and single-qubit gates reducing to identity.

B. Pulse-Level Compilation

Physical quantum gates are implemented through microwave pulses. We model pulse optimization using the Lindblad master equation [6]:

$$\frac{d\rho}{dt} = -i[\mathcal{H}(t), \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right) \quad (1)$$

where $\mathcal{H}(t)$ is the time-dependent control Hamiltonian, L_k are Lindblad operators, and γ_k are rates derived from T_1 and T_2 times.

C. IQM Garnet Architecture

The IQM Garnet processor [7] features 20 superconducting transmon qubits with $T_1 = 37 \mu\text{s}$, $T_2 = 9.6 \mu\text{s}$, single-qubit gate error 0.1%, and two-qubit gate error 0.6%.

TABLE I
PER-PASS EFFECTIVENESS METRICS.

Pass	Gates Removed	% Improved	Rank
cancel	14,024	68%	1
rotate	6,512	29%	2
identity	55	9%	3
commute	0	0%	4

III. SYSTEM ARCHITECTURE

Our framework comprises five pipeline stages:

Stage 1: Parse & Validate. Input circuits in OpenQASM 3.0 format are parsed and validated.

Stage 2: Optimize. The circuit is passed to the C++ optimizer via subprocess, applying configurable optimization passes.

Stage 3: Route. SABRE routing [8] maps logical qubits to physical qubits, inserting SWAP gates to satisfy connectivity constraints.

Stage 4: Pulse Compile. Native gates are compiled to microwave pulse sequences (20 ns for single-qubit, 40 ns for two-qubit gates).

Stage 5: Simulate. The pulse sequence is simulated under a Lindblad decoherence model. Process fidelity $\mathcal{F}_{\text{proc}}$ and state fidelity $\mathcal{F}_{\text{state}}$ are computed.

IV. METHODOLOGY

A. Circuit Corpus

We evaluate on a diverse corpus of 371 circuits: GHZ states (2–12 qubits), QFT (2–8 qubits), QAOA for MaxCut, and random circuits (5–30 layers, 4–8 qubits).

B. Fidelity Metrics

Process fidelity measures overlap between implemented and target unitaries:

$$\mathcal{F}_{\text{proc}} = \frac{|\text{Tr}(U_{\text{target}}^\dagger U_{\text{impl}})|^2}{d^2} \quad (2)$$

State fidelity for target state $|\psi\rangle$ and output density matrix ρ :

$$\mathcal{F}_{\text{state}} = \langle \psi | \rho | \psi \rangle \quad (3)$$

V. SIMULATION RESULTS

A. Overall Performance

Across 371 circuits: mean process fidelity 0.680 ± 0.224 , median 0.718, mean gate reduction 23.1%, maximum reduction 96.2%.

B. Pass Effectiveness

Gate cancellation emerges as the most impactful, eliminating 14,024 gates with 68% of circuits showing improvement. Table I summarizes per-pass metrics.

The optimal pass ordering is `cancel` $\rightarrow$ `commute` $\rightarrow$ `rotate`.

TABLE II
HARDWARE FIDELITY MEASUREMENTS ON IQM RESONANCE GARNET.

Circuit	Gates	Reduction	Fid (Orig)	Fid (Opt)
ghz_4q	4 $\rightarrow$ 4	0%	0.494	0.469
ghz_8q	8 $\rightarrow$ 8	0%	0.406	0.375
ghz_12q	12 $\rightarrow$ 12	0%	0.256	0.288
qft_4q	30 $\rightarrow$ 9	70%	0.100	0.088

C. Scaling Analysis

Strong correlations exist between circuit parameters and process fidelity. Pulse duration exhibits the strongest predictive power ($r = -0.743$, $R^2 = 0.55$), highlighting decoherence as the dominant fidelity-limiting factor.

VI. HARDWARE VALIDATION

We validated findings on the IQM Resonance Garnet 20-qubit processor with 8 quantum jobs (160 shots each, 100% success rate).

Key findings: (1) The optimizer correctly identifies GHZ circuits as already minimal. (2) QFT achieved 70% gate reduction (30 $\rightarrow$ 9 gates). (3) Hardware fidelity decreases with qubit count, consistent with NISQ behavior.

VII. DISCUSSION

Our results provide actionable guidance for quantum compiler design:

- **Prioritize cancellation:** Gate cancellation provides the largest fidelity gains (14,024 gates eliminated).
- **Commutation enables cancellation:** While commutation provides no direct gate reduction, it creates opportunities for subsequent cancellation.
- **Minimize pulse duration:** The strong correlation ($r = -0.74$) emphasizes decoherence-aware optimization.

VIII. CONCLUSION

We have presented qco-integration, a framework for end-to-end fidelity analysis with hardware validation on IQM Resonance Garnet. Gate cancellation is the most effective pass (68% improvement rate), pulse duration strongly predicts fidelity ($R^2 = 0.55$), and the optimal sequence is `cancel` $\rightarrow$ `commute` $\rightarrow$ `rotate`. Hardware validation confirms the optimizer produces valid executable circuits with 70% gate reduction on QFT.

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